ZTID-IoV: Zero-Trust Intrusion Detection in IoV Using Neurosymbolic AI Approach With Federated Meta-Learning

Farhan Ullah, Gautam Srivastava, *Senior Member, IEEE*, Leonardo Mostarda, *Member, IEEE*, and Umar Raza, *Senior Member, IEEE*

***Abstract*— The rapid growth of the Internet of Vehicles (IoVs) and smart consumer electronics has generated cybersecurity concerns that require an intelligent, adaptable, and privacy-preserving Intrusion Detection System (IDS). This study introduces ZTID-IoV, a novel neurosymbolic AI framework that integrates federated learning, lightweight transformers, and meta-learning to improve threat detection while preserving user privacy in consumer IoVs. Our approach leverages neural components such as a transformer model for recognizing patterns in network traffic, combined with symbolic AI techniques such as self-organizing maps for interpretable client clustering and rule-guided reasoning, to achieve robust cybersecurity in distributed environments. A lightweight transformer architecture optimizes performance for resource-constrained edge devices, and SOM-based clustering enhances model aggregation by grouping devices with similar behavioral patterns. The proposed system employs Model-Agnostic Meta-Learning (MAML) to enable rapid adaptation to emerging threats across diverse consumer devices, while federated learning ensures decentralized model training without exposing sensitive user data. Experiments on real-world IoT intrusion datasets demonstrate that our framework achieves higher detection accuracy compared to centralized and pure neural approaches while maintaining low computational overhead. Additionally, the neurosymbolic design provides interpretable threat explanations, crucial for consumer applications where transparency is essential. The results highlight the potential of ZTID-IoV in enabling zero-trust security for IoV and other connected consumer electronics. This work contributes to the evolving landscape of AI-driven cybersecurity by addressing critical challenges in privacy and adaptability, making ZTID-IoV particularly suitable for next-generation IoV ecosystems.**

***Index Terms*— Neurosymbolic artificial intelligence (AI), IoV security, zero-trust IDS, federated learning, model-agnostic meta-learning, self-organizing maps.**

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I. Introduction

The rapid convergence of IoVs and smart consumer electronics has resulted in the development of an interconnected transportation ecosystem that is revolutionizing in-vehicle experiences, vehicle automation, and mobility services. These developments improve traffic management, road safety, and user comfort, but also pose cybersecurity risks to key infrastructure, user privacy, and vehicle operations [1]. Modern connected vehicles exchange sensitive data with roadside infrastructure, cloud platforms, and consumer electronics, creating a complex attack surface. The real-time and privacy requirements of consumer applications and transportation systems are supported by advanced security solutions, which are designed to safeguard networks [2].

Centralized machine learning techniques are used in traditional intrusion detection to gather unstructured network traffic from smart devices and vehicles. This centralized solution exposes sensitive driver and vehicle operational data, which raises privacy concerns. Moreover, the presence of a single aggregation point introduces system vulnerabilities, where a breach could lead to widespread failures. Federated learning offers a potential alternative by reducing data centralization and enabling remote model training. Its implementation in the IoV may involve three major challenges. First, the current implementations are insufficiently adaptable to the changing attack vectors across a variety of consumer electronics and automotive platforms. Second, they frequently fall short of the transparent decision-making requirements necessary for vehicle safety certification. Finally, federated learning is vulnerable to model poisoning and other adversarial attacks targeting shared parameters [3], [4].

The cybersecurity of IoV-consumer ecosystems is facing an increasing number of challenges as a result of their unique characteristics. A variety of data patterns are generated by interdependent parts, such as entertainment systems, personal smart gadgets, and vehicle control units. These are challenging for conventional detection systems to effectively process, as demonstrated by malicious applications that target infotainment interfaces and spoofed attacks on tire pressure monitors [5]. Conventional detection systems struggle to analyze data streams from connected devices such as automotive ECUs, smartphones, and infotainment systems due to their diversity. Automotive safety systems need extremely low latency responses, whereas consumer applications need effective and easily understood security measures. Moreover, attack methods against automotive and consumer networks are always changing. This needs adaptive security systems that can detect emerging threats without costly retraining or threatening user privacy via centralized data collection [6]. Current solutions are inadequate in their ability to effectively address the multifaceted security challenges that IoV-consumer ecosystems present. Although neural network-based methods are highly effective at recognizing patterns, they frequently lack transparency and provide limited insight into their decision-making processes. In contrast, rule-based systems are more interpretable; however, they are unable to respond to new threats or manage the complexity of multimodal data produced by contemporary interconnected devices. Furthermore, many existing techniques treat vehicle and consumer device security separately, ignoring valuable connections that result from their combined behavior. A unified security architecture that combines machine learning and symbolic reasoning while protecting user privacy is needed to address these challenges. This paper introduces ZTID-IoV, a zero-trust IDS framework that is specifically designed for IoV and smart consumer environments. This framework is founded on a novel neurosymbolic AI paradigm. This paper adds four significant contributions to the domain:

1) We present ZTID-IoV, a neurosymbolic AI framework that uses federated learning, lightweight transformers, and meta-learning to improve threat detection and user privacy. The framework combines transformers for processing multivariate data with SOM for interpretable threat clustering of automotive systems and consumer electronics.
2) This strategy supports user privacy while preserving threat detection capabilities. MAML-based federated meta-learning enables the rapid adaptation of vehicle models to novel attacks. The method ensures data privacy and promptly identifies evolving risks from limited instances, making it helpful against novel in-vehicle system and consumer application attacks.
3) The proposed study adopts a zero-trust approach to secure the federated learning process for intrusion detection across varied clients in the IoVs. To improve robustness, cluster-aware aggregation is used, in which clients are grouped according to behavioral similarities using SOM analysis. This system maintains rigid trust boundaries, continuously checks client behavior, mitigates poisoning risks, and reduces communication overhead, following zero-trust principles. This strategy is critical in IoV, as frequent vehicle-to-vehicle and vehicle-to-infrastructure interactions create many attack surfaces.
4) The proposed approach is rigorously assessed on four standard datasets with differing client counts. This evaluation emphasizes the framework's capacity to sustain robustness, generalize, and scale effectively in a variety of real-world scenarios.

The remainder of the paper is organized as follows. Related work is reviewed in Section II, the proposed scheme is presented in Section III, the results and analysis are discussed in Section IV, and the study is concluded in Section V.

II. Related Work

Several research studies utilize SOM, MAML, and federated learning to enhance the efficiency and accuracy of intrusion detection systems [1], [6], [7], [8]. The following sections examine various approaches that utilize FL, as well as the combined application of SOM and MAML, to enhance cyberattack classification in distributed environments.

A. SOM/MAML Methods

The Deep Autoencoding Gaussian Mixture Model (DAGMM) has demonstrated superior performance in comparison to conventional two-stage methods. However, it is unable to maintain the input topology as a result of the constraints imposed by its autoencoder design. Chen et al. [9] proposed an SOM-assisted DAGMM to address this issue. The model improves the accuracy of intrusion detection by maintaining the topological structure and improving the representation of features through SOM integration. Almi'ani et al. [10] proposed an IDS that is based on a clustered SOM. The system functions in two phases: initially, a SOM is constructed, and subsequently, its neurons are hierarchically clustered using k-means. This solution tackles connection record sensitivity and processing time issues. Lu et al. [8] introduced an IoT-based IDS that is specifically designed for scenarios with restricted training samples. The approach uses meta-learning and a two-layer MAML-based training strategy to train a generalized model on malicious traffic data from various attack types. Wang et al. [11] utilized online meta-learning to design a network IDS that includes anti-forgetting mechanisms and few-shot learning capabilities. It learns efficiently with minimal input and prevents major errors by saving and reapplying previous models. Gong et al. [12] created a few-shot learning-based IDS to overcome challenges in evaluating similar video frames. The method uses a trained augmented MAML framework on original video frames and segmented track area masks. The goal of this work is to enhance training efficiency by providing theoretical insight and practical solutions for handling frame similarity during meta-learning. The strategy improves typical supervised learning models in intrusion detection and adaptation with few samples in real-world railway video data simulations. Hu et al. [13] proposed a privacy-preserving few-shot traffic detection approach based on federated meta-learning. Clients can respond to local threats while the server collects global data using dual-phase updates. This method considers enhanced persistent threat detection as a model generalization effort. This technique detects advanced persistent threats with good accuracy, low latency, and robust adaptability across domains.

B. FL-Based Methods

Mothukuri et al. [14] proposed a federated learning-based solution for detecting anomalies using local device data.

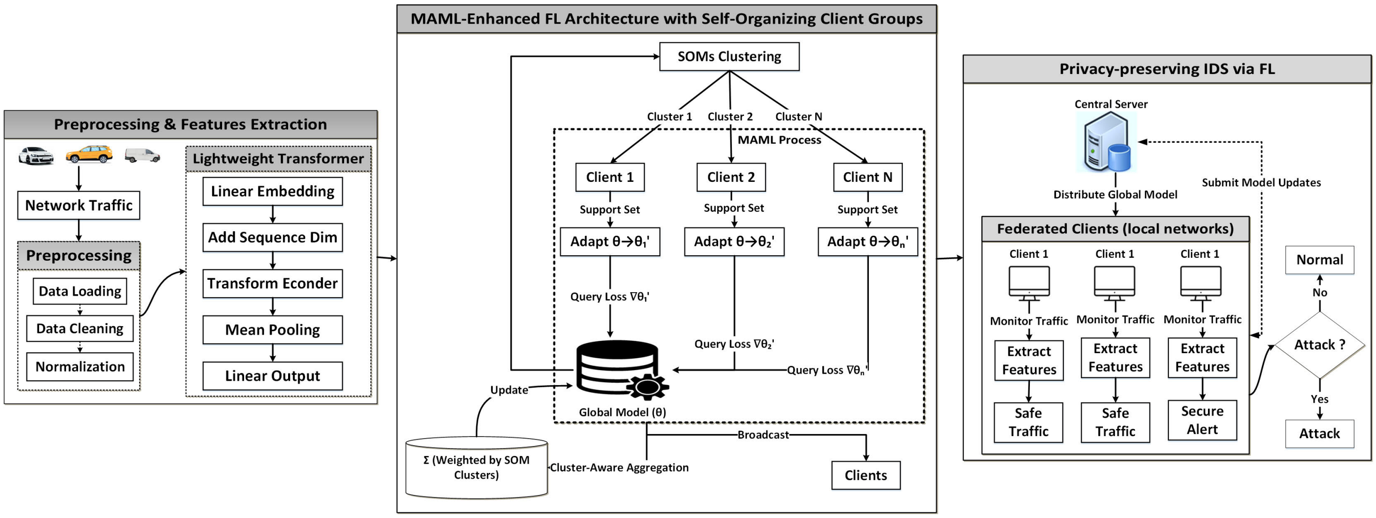


Fig. 1. ZTID-IoV: Detecting Zero-Trust Intrusions in the Internet of Vehicles with a Neurosymbolic AI Approach and Federated Meta-Learning.

Gated recurrent units are incorporated into the model, and only model parameters are sent to a central server. The ensemble method improves privacy protection and detection over centralized methods. Huang et al. [15] proposed a federated-based IDS for the IoT that prioritizes data privacy across edge devices. This technique is adequate to low-resource settings because it uses a lightweight MobileNet-Tiny model for feature extraction and turns network traffic into visuals. Evaluation results indicate robust detection capabilities, regardless of the presence of imbalanced data. Chen et al. [16] proposed privacy score and evasion rate measures for evaluating federated learning-based intrusion detection, focusing on adversarial impact and traffic similarity. The Kitsune model can be circumvented by fraudulent communication, as their research demonstrated. To address this, they devised an optimization method that improves privacy and detection by decreasing gradient similarity and boosting input variety. The unequal distribution of IoT data limits edge-based intrusion detection, which leads to a high rate of false alarms for unknown attacks, as highlighted by Lu et al. [17]. To eliminate manual labeling, they suggested self-labeling using a cosine similarity-based feature extractor and selective model aggregation. The technique demonstrated high performance on the CICIDS2017 dataset, even in the absence of human labeling. Uddin et al. [18] identified significant problems in cross-domain federated learning, including differences in data properties, labeling, and security needs. Transfer learning, including fine-tuning global models with local domain data, improves adaptability. Meta-learning techniques, such as MAML, allow models to adjust quickly with minimal data. Domain adaptation techniques and federated transfer learning are also proposed for improved generalization while preserving privacy across several domains. Yan et al. [19] improved anomaly-based intrusion detection with federated learning in response to growing cyber risks in IoT. Although existing systems promote global performance, they frequently overlook the adaptation of local models. Participant models can be customized to local data using federated meta-learning. The utilization of clustered gradient similarities to select clients expedites convergence and enhances real-time intrusion detection.

Our novel ZTID-IoV method detects IoT threats while preserving data by utilizing lightweight transformers, meta-learning, and federated learning. The method uses interpretable device aggregation using SOMs, symbolic reasoning, and transformer models to identify traffic patterns. A lightweight transformer improves edge device efficiency, while SOM-based clustering facilitates model aggregation. ZTID-IoV responds swiftly to new threats via MAML, and federated learning allows for secure, decentralized model upgrades without releasing sensitive data.

III. ZTID-IoV: Federated Neurosymbolic Zero-Trust Intrusion Detection in IoV

The proposed ZTID-IoV approach is shown in Figure 1. This approach integrates meta-learning, federated learning, and self-organizing maps to develop a cybersecurity solution that is adaptive and maintains privacy in IoV environments. Effective federated aggregation can be implemented by features such as MAML, which enables rapid adaptation to new attack patterns, a lightweight detection model based on transformers, and SOM-based client clusterings.

A. Dataset Preparation

We evaluate the suggested method on four standard datasets. The CICIOTDIAD2024[1] [20] is a comprehensive dataset of IoT attacks developed by the Canadian Institute for Cybersecurity. It comprises 33 attacks against 105 IoT devices to improve the identification of devices and the detection of anomalies. To address limitations of unreliable IDS datasets, the second dataset, CICIDS2017,[2] comprises realistic benign traffic and several recent attacks [21]. It provides labeled PCAPs and

[1] https://www.unb.ca/cic/datasets/iot-diad-2024.html
[2] https://www.unb.ca/cic/datasets/ids-2017.html

TABLE I
CLASS DISTRIBUTION ACROSS ALL DATASETS

Dataset & Class	**Samples**	**Percentage**	**Ratio**
CICIOTDIAD2024			
Benign	38,896	41.6%	1:1.4
Attack	54,401	58.4%	1:0.7
Total	93,297	100%	–
CIC-IDS2017			
DDoS	5,847	42.5%	1:2.7
Port Scan	5,692	41.4%	1:2.8
Web Attacks	2,180	15.8%	1:7.2
Infiltration	36	0.3%	1:437.6
Total Attacks	13,755	100%	–
CICIoV2024			
Benign	1,223,737	89.4%	1:0.12
DoS	74,663	5.5%	1:18.3
Spoofing: GAS	9,991	0.7%	1:136.8
Spoofing: Str. Wheel	19,977	1.5%	1:68.4
Spoofing: Speed	24,951	1.8%	1:54.8
Spoofing: RPM	54,900	4.0%	1:24.9
Total	1,368,219	100%	–
UAVAttack			
Normal	768,666	97.7%	1:0.02
Anomaly	18,162	2.3%	1:42.3
Total	786,828	100%	–

flow-based CSV files for accurate IDS evaluation by imitating human behavior across protocols with the B-Profile system. A variety of devices, sensors, communication protocols, and edge configurations are utilized in the development of the third dataset, CICIOV2024[3] [22]. This testbed is part of the IoT and Industrial IoT ecosystem. The fourth dataset, UAVAttack[4] [23], contains logs from simulated and real UAV flights under benign and attack situations. It highlights the challenges of GPS spoofing and interference, which are difficult to replicate in a variety of research environments. Table I summarizes network traffic distributions from four datasets. The "Dataset & Class" field identifies each dataset as well as its traffic categories. The terms "Samples" and "Percentage" indicate the absolute traffic counts and the relative frequency of each class in the dataset, respectively. The "Ratio" column compares class ratios to other classes' total samples in the dataset.

B. Data Preprocessing

Data quality and consistency are both improved during the extensive preparation phase that the network traffic data goes through. The integrity of the dataset is first ensured by removing any columns with non-numeric values, missing entries, or infinite values. The dimensionality is carefully maintained by retaining only the relevant features that are required for the learning task, as network traffic frequently contains a high number of features. To normalize input ranges, min-max normalization is used to transform all feature values to the [0,1] range, which improves convergence during training. Label encoding is the process of converting categorical attack types into numerical labels that may be employed with machine learning models. Subsampling is applied only when the dataset exceeds a predefined size limit max_samples. Stratified random sampling is implemented to ensure reproducibility and preserve class distributions, with a fixed random seed random_state = 42. Importantly, each experiment employed the same sampling method. The same randomized selection method is consistently applied to either complete or subsampled datasets. The data is partitioned among client nodes to simulate federated environments following preprocessing, and an independent global test set is maintained for the final evaluation. A training-to-testing split of 80-20 ensures reliable model validation [24].

[3]https://www.unb.ca/cic/datasets/iov-dataset-2024.html
[4]https://ieee-dataport.org/open-access/uav-attack-dataset

C. Lightweight Transformer Architecture

The intrusion detection model uses an improved transformer architecture that is designed to analyze tabular data efficiently. Equation 1 shows that the input features $\mathbf{X} \in \mathbb{R}^{n \times d_{in}}$ are first translated into a lower-dimensional token space by a linear projection [25].

$$\mathbf{Z}_0 = \mathbf{X}\mathbf{W}_e + \mathbf{b}e \quad \text{where} \quad \mathbf{W}e \in \mathbb{R}^{din \times dtoken}, \mathbf{b}e \in \mathbb{R}^{dtoken} \tag{1}$$

The compact embedding dimension is denoted by $d_{token} = 32$ in this case. The $h = 4$ attention units and $L = 2$ layers of multi-head self-attention are utilized by the transformer encoder. Equations 2 and 3 determine how each layer l computes.

$$\mathrm{MHSA}(\mathbf{Z}_l) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^O \tag{2}$$

$$\mathrm{head}_i = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}}\right)\mathbf{V}_i \tag{3}$$

$\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}i$ represent learnt query, key, and value projections for head i, and $d_k = dtoken/h$. Equation 4 illustrates how layer outputs include residual connections and layer normalization.

$$\begin{aligned}\mathbf{Z}_{l+1} = \mathrm{LayerNorm}(&\mathbf{Z}_l + \mathrm{FFN}(\mathrm{LayerNorm}(\mathbf{Z}_l \\ &+ \mathrm{MHSA}(\mathbf{Z}_l))))\end{aligned} \tag{4}$$

The classification head implements linear projection subsequent to global average pooling, as illustrated in equation 5.

$$\mathbf{y} = \mathrm{Softmax}\left(\mathrm{Linear}\left(\frac{1}{n}\sum_{i=1}^{n}\mathbf{Z}_L^{(i)}\right)\right) \tag{5}$$

The efficient lightweight transformer features a lower token size ($d_{token} = 32$), fewer attention heads ($h = 4$), and a shallow depth of only two layers ($L = 2$). These improvements lower computational and memory overhead, making the model suitable for real-time intrusion detection in resource-constrained environments. Despite its modest size, it may capture complex feature interactions in tabular network traffic data, enhancing the overall effectiveness of the proposed technique.

D. Privacy-Preserving Federated Learning

The proposed method adopts a privacy-preserving federated learning framework. A global model Θ_G is cooperatively trained by K distributed clients without exchanging raw data. Each client k keeps its local dataset $\mathcal{D}_k = (\mathbf{x}i, y_i)i = 1^{n_k}$ and updates the local model using stochastic gradient descent, as indicated in equation 6. Only model parameters are transferred to the central server, preserving data privacy throughout the training process [26], [27].

$$\Theta_k^{(t+1)} = \Theta_k^{(t)} - \eta \nabla \mathcal{L}(\Theta_k^{(t)}; \mathcal{B}_k) \tag{6}$$

A minibatch is indicated by $\mathcal{B}_k \subset \mathcal{D}_k$, and the learning rate is η. Equation 7 shows that clients have the choice to use MAML-based adaptation.

$$\Theta_k' = \Theta_G - \alpha \nabla \mathcal{L}(\Theta_G; \mathcal{D}_k^{\text{supp}}) \tag{7}$$

The support set $\mathcal{D}k^{\text{supp}}$ and the adaptation rate α are being utilized. The servers use SOM to group clients into groups according to their data distributions $\mu_k k = 1^K$, where μ_k is the mean feature vector of $\mathcal{D}_k$. The SOM uses competitive learning to create cluster allocations $c(k) \in 1, \ldots, C$, as illustrated in equation 8.

$$c(k) = \arg\min_j \|\mu_k - \mathbf{w}_j\| \tag{8}$$

Cluster-aware aggregation computes weighted averages for SOM weight vectors $\mathbf{w}jj = 1^C$ using equation 9.

$$\Theta G^{(r+1)} = \sum c = 1^C \frac{|S_c|}{K} \left(\frac{1}{|S_c|} \sum_{k \in S_c} \Theta_k \right) \tag{9}$$

The clients in cluster c are denoted by $S_c = k : c(k) = c$. Convergence is evaluated by the global validation loss, as illustrated in equation 10, and the federated rounds $r \in 1, \ldots, R$ proceed until convergence.

$$\frac{|\mathcal{L}\text{val}^{(r)} - \mathcal{L}\text{val}^{(r-1)}|}{\mathcal{L}_{\text{val}}^{(r-1)}} < \epsilon \tag{10}$$

The convergence threshold is denoted by the symbol ϵ. This architecture preserves privacy while allowing for customized detection through client-specific adaptation and effective knowledge transfer via clustered aggregates. The current architecture ensures privacy by utilizing federated learning, in which clients never exchange raw data but only model parameters.

E. Meta-Learning Integration

Client data is typically non-IID in federated learning environments, resulting in substantial variations in local model performance. This approach integrates MAML into the federated learning architecture to promote generalization across varied client tasks. Meta-learning learns a shared initialization that can be readily adjusted to new client-specific tasks with minimum data computation. This can be helpful in federated environments where data heterogeneity and limited communication cycles prevent global convergence. The MAML framework uses a two-stage optimization procedure to increase the model's flexibility [13]. The inner loop implements client-specific fine-tuning on support sets $\mathcal{D}_k^{\text{supp}}$ for every client task $\mathcal{T}_k \sim p(\mathcal{T})$, as illustrated in equation 11.

$$\Theta_k' = \Theta - \alpha \nabla_\Theta \mathcal{L}_{\mathcal{T}_k}(\Theta; \mathcal{D}_k^{\text{supp}}) \tag{11}$$

The inner loop learning rate is denoted by α in this scenario. The meta-update optimizes the starting attributes across all tasks by gradient descent on query sets $\mathcal{D}_k^{\text{query}}$, as shown in equation 12 [28].

$$\Theta \leftarrow \Theta - \beta \nabla_\Theta \sum_{\mathcal{T}_k \sim p(\mathcal{T})} \mathcal{L}_{\mathcal{T}_k}(\Theta_k'; \mathcal{D}_k^{\text{query}}) \tag{12}$$

It uses β as its meta-learning rate. The inner-loop adaptation (Eq. 11) is conducted locally on each client during each federated round, followed by query loss broadcast to the server. During each federated round, the server does the meta-update (Eq. 12) before the cluster-aware aggregation step in Algorithm 1.

This ensures that the global initialization Θ_G is refined each round while still benefiting from the clustering-based aggregation. Equation 13 demonstrates that this formulation allows for rapid adaptation to novel attack patterns $\mathcal{T}_{\text{new}}$ with few samples m.

$$\Theta_{\text{new}}' = \Theta - \alpha \nabla_\Theta \mathcal{L}_{\mathcal{T}_{\text{new}}}(\Theta; \mathcal{D}_{\text{new}}^m) \tag{13}$$

The meta-objective maintains global model generalization by reducing anticipated loss across tasks as shown in equation 14.

$$\min_\Theta \mathbb{E}_{\mathcal{T}_k \sim p(\mathcal{T})}[\mathcal{L}_{\mathcal{T}_k}(\Theta_k'; \mathcal{D}_k^{\text{query}})] \tag{14}$$

This bi-level optimization rapidly customizes new client environments and generalizes across varied intrusion patterns. Thus, incorporating meta-learning into the federated setup is critical for improving model adaptability, scalability, and robustness in dynamic and heterogeneous intrusion detection scenarios.

F. SOM-Based Client Clustering

The SOM originally proposed by Kohonen [29] uses an unsupervised neural architecture to project high-dimensional client data distributions $\mu_k{}_{k=1}^K \subset \mathbb{R}^d$ onto a two-dimensional grid $\mathcal{G} = \{1, \ldots, M\} \times \{1, \ldots, N\}$. This allows for intelligent client grouping [30]. According to equation 15 and Figure 2, the Kohonen algorithm employs competitive learning to maintain a prototype vector $\mathbf{w}_{ij} \in \mathbb{R}^d$ for every grid node $(i, j) \in \mathcal{G}$.

$$(i^*, j^*) = \arg\min_{(i,j)\in\mathcal{G}} \|\boldsymbol{\mu}_k - \mathbf{w}_{ij}\|_2 \tag{15}$$

The SOM preserves topological relationships across neighborhood alterations, as shown in equation 16. Equation 17 shows the relationship between the Gaussian neighborhood function $h(\cdot)$ and the decaying learning rate $\eta(t)$.

$$\mathbf{w}ij \leftarrow \mathbf{w}ij + \eta(t) \cdot h((i, j), (i^\cdot j^\rangle; t) \cdot (\mu k - \mathbf{w}ij) \tag{16}$$

$$h((i, j), (i^\cdot j^\rangle; t) = \exp\left(-\frac{|(i, j) - (i^\cdot j^\rangle|^2}{2\sigma^2(t)} \right) \tag{17}$$

This leads to cluster assignments $c(k) = (i^\cdot j^\rangle$, which enable cluster-aware model aggregation, as demonstrated in equation 18. Equation 19 demonstrates that the final global model incorporates updates particular to each cluster.

$$\Theta G^{(c)} = \frac{1}{|S_c|} \sum k \in S_c \Theta_k, \quad S_c = k \mid c(k) = c \tag{18}$$

$$\Theta G = \sum c \in \mathcal{C} \frac{|S_c|}{K} \Theta_G^{(c)} \tag{19}$$

Algorithm 1 ZTID-IoV: Zero-Trust Intrusion Detection for Internet of Vehicles

Input: Client datasets: $\{\mathcal{D}_k\}_{k=1}^{K}$, Attack classes $\mathcal{C}$
Output: Global intrusion detection model Θ_G

Function `ZTID-IoV(`$\{\mathcal{D}_k\}, \mathcal{C}$`)`**:**
- **Initialization:** Initialize global transformer model Θ_G with $d_{token} = 32$, $h = 4$, $L = 2$;
- Initialize SOM grid $\mathcal{G} = \{1, \ldots, M\} \times \{1, \ldots, N\}$ with random $\mathbf{w}_{ij}$;
- **Data Preparation: for** *each client* k **do**
  - Clean $\mathcal{D}_k$: remove non-numeric, NaN, $\pm\infty$ values;
  - Normalize: $\hat{x}_{ij} = \frac{x_{ij} - \min(\mathbf{x}_j)}{\max(\mathbf{x}_j) - \min(\mathbf{x}_j)}$;
  - Encode labels: $y \leftarrow \text{LabelEncoder}(\mathcal{C})$;
  - Split $\mathcal{D}_k$ into support/query sets: $\mathcal{D}_k^{\text{supp}}$, $\mathcal{D}_k^{\text{query}}$;
- **for** *each federated round* $r = 1$ *to* R **do**
  - **Client Clustering: for** *each client* k **do**
    - Compute mean feature vector $\mu_k = \mathbb{E}[\mathcal{D}_k]$;
    - Find BMU: $(i^*, j^*) = \arg\min_{(i,j)\in\mathcal{G}} \|\mu_k - \mathbf{w}_{ij}\|_2$;
    - Update SOM: $\mathbf{w}_{ij} \leftarrow \mathbf{w}_{ij} + \eta(t)h((i,j),(i^*,j^*);t)(\mu_k - \mathbf{w}_{ij})$;
    - Assign cluster: $c(k) = (i^*, j^*)$;
  - **Client Training: for** *each client* k *in parallel* **do**
    - **if** *MAML enabled* **then**
      - Inner-loop adaptation on client k;
      - Inner loop: $\Theta'_k = \Theta_G - \alpha\nabla\mathcal{L}(\Theta_G; \mathcal{D}_k^{\text{supp}})$;
      - Compute query loss: $\mathcal{L}_k = \mathcal{L}(\Theta'_k; \mathcal{D}_k^{\text{query}})$;
    - **else**
      - Local SGD: $\Theta_k \leftarrow \Theta_G - \eta\nabla\mathcal{L}(\Theta_G; \mathcal{B}_k)$;
    - Send Θ_k to server;
  - **Meta-Update (Outer Loop):** $\Theta_G \leftarrow \Theta_G - \beta\nabla_{\Theta_G}\sum_{k=1}^{K}\mathcal{L}_k$ **;** Executed once per federated round, on the server, before aggregation
  - **Cluster-Aware Aggregation: for** *each cluster* $c \in \mathcal{G}$ **do**
    - $\Theta_G^{(c)} \leftarrow \frac{1}{|S_c|}\sum_{k\in S_c}\Theta_k$;
  - Update global model: $\Theta_G \leftarrow \sum_c \frac{|S_c|}{K}\Theta_G^{(c)}$;
  - **Convergence Check: if** $\frac{|\mathcal{L}_{val}^{(r)} - \mathcal{L}_{val}^{(r-1)}|}{\mathcal{L}_{val}^{(r-1)}} < \epsilon$ **then**
    - Break;
- **Output:** Return Θ_G for intrusion detection;

The identified clusters are denoted as $\mathcal{C}$. This technique improves model accuracy on heterogeneous data by using unique cluster-wise updates and accelerates convergence by $\mathcal{O}(\sqrt{K/C})$ compared to ordinary federated averaging [7]. Initializing the SOM grid dimensions $M \times N$ based on client population K provides a default of 25 nodes. To maintain topology, we use the heuristic $C = \lfloor\sqrt{5K}\rfloor$ to compute the number of clusters on a near-square grid ($M \approx N$). Initializing prototype vectors $\mathbf{w}_{ij}$ through PCA sampling of client data means $\{\boldsymbol{\mu}_k\}$, while neighborhood radii and learning rates follow exponential decay schedules. This initialization strategy guarantees the construction of robust clusters while simultaneously preserving computational efficiency. Variations in client participation during federated rounds are effectively handled by the adaptive nature of our SOM implementation. The SOM-based clustering approach operates by dynamically projecting only active clients into the SOM grid. Each active client is assigned to the best-matching node based on data distribution, without requiring full client participation. Competitive learning updates pertinent nodes in the SOM incrementally, preserving significant topological relationships. This ensures a resilient and adaptable model updates even with fluctuating client numbers by aggregating available clients within each cluster.

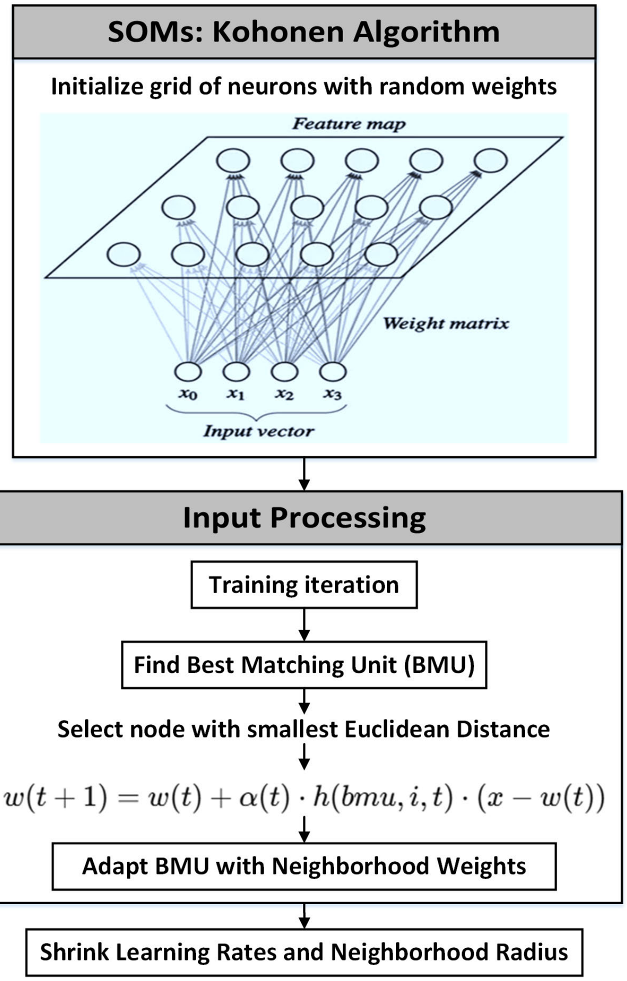
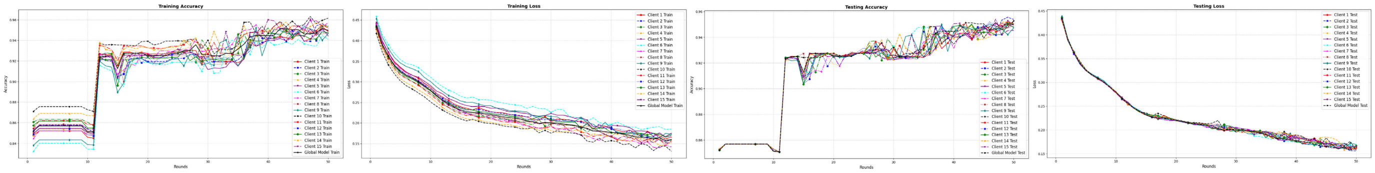


Fig. 2. Client clustering using Kohonen SOMs to group similar IoV devices.

IV. Results and Discussions

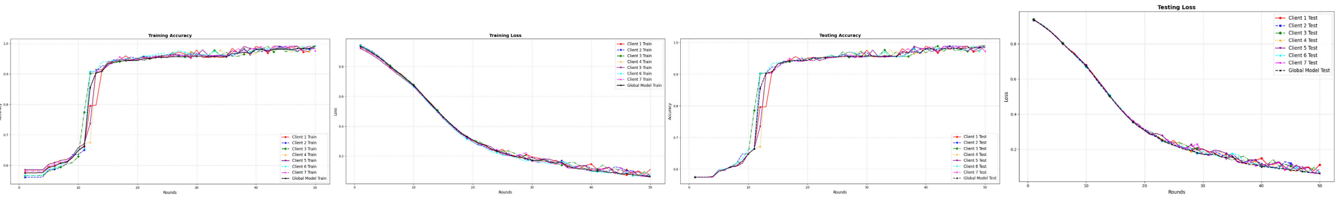
A. Performance Indicators

The ZTID-IoV architecture is evaluated across multiple distributed clients to validate its effectiveness and robustness



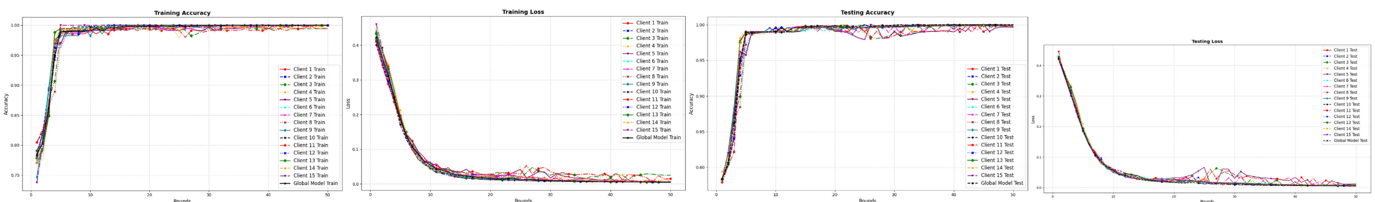
(a) Training Accuracy (b) Training Loss (c) Testing Accuracy (d) Testing Loss

Fig. 3. Dynamic accuracy/loss curves with CICIoTDIAD2024 dataset using 15 clients.



(a) Training Accuracy (b) Training Loss (c) Testing Accuracy (d) Testing Loss

Fig. 4. Dynamic accuracy/loss curves with CICIoV2024 dataset using 7 clients.



(a) Training Accuracy (b) Training Loss (c) Testing Accuracy (d) Testing Loss

Fig. 5. Jamming: Dynamic accuracy/loss curves with UAVAttack dataset using 15 clients.

in real-world IoT and IoV environments. Equations 20 to 23 define the core classification outcomes: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN), which are used to compute key performance metrics including precision, recall, F1-score, and accuracy. Precision is the proportion of intrusions that are correctly identified, while recall is indicative of the capacity to identify actual intrusions. The F1-score balances both, while accuracy reflects total prediction accuracy.

$$Precision = \frac{TP}{(TP + FP)} \tag{20}$$

$$Recall = \frac{FP}{(FP + TN)} \tag{21}$$

$$F1 - score = \frac{(2 * TP)}{(2TP + FP + FN)} \tag{22}$$

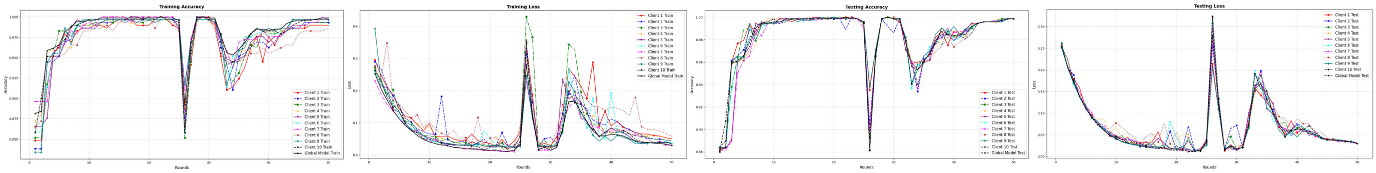
$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{23}$$

B. Performance Analysis and Comparisons

Figure 3 displays the performance of the proposed method, comparing epoch loss and accuracy for various clients. Some clients outperform others, with client 10 achieving 96.26% accuracy in Round 49 and client 14 reaching 95.51% in Round 34, while client 6 lags with 82.64% accuracy in Round 7, likely due to data or resource disparities. Figure 4 displays client training performance, including epoch-wise loss and accuracy. Initially, the models demonstrate low accuracy (55–60%) and a high loss (0.9–1.0). clients 4 and 5 demonstrate the most rapid rate of improvement. Particularly for clients 3 and 6, there is a substantial decrease in loss (0.3–0.5) and an increase in accuracy (90–95%) between Rounds 10–20. By the final stages, all clients achieve high accuracy (99%) and low loss (0.06). Figure 5 displays the training performance of 15 clients over 50 federated learning rounds, with changing loss and accuracy. Most clients initially have moderate accuracy (75-80%) and higher loss values (0.5-0.6). Early rounds show increased accuracy, particularly after Round 6, with several clients above 95% accuracy. By the 20th round, almost all clients achieve convergence with a loss of less than 0.01-0.02 and an accuracy of 99-100%.

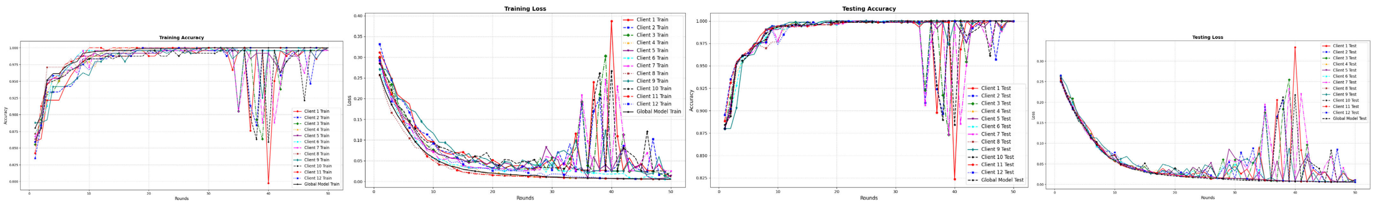
The client-wise training performance over 50 rounds is presented in Figure 6. Clients initially exhibit greater loss values (0.3–0.4) and moderate accuracy (80–90%). The majority of clients achieve an accuracy rate of approximately 99% and a loss rate of approximately 0.01–0.02 by the 15th round. Figure 7 illustrates that clients experienced a rapid improvement in accuracy and loss, rising from ∼80–90% accuracy with a high loss (∼0.4) to ≥99% accuracy and loss <0.01 by Round 15. During the final rounds, all clients achieved 100% accuracy with minimum loss, indicating strong convergence.

The performance parameters for the proposed method employing the CICIoTDIAD2024 dataset for 10 clients are shown in Table II. The model had high accuracy (95.59%), with good precision, recall, and F1-scores for both benign



(a) Training Accuracy (b) Training Loss (c) Testing Accuracy (d) Testing Loss

Fig. 6. Spoofing: Dynamic accuracy/loss curves with UAVAttack dataset using 10 clients.



(a) Training Accuracy (b) Training Loss (c) Testing Accuracy (d) Testing Loss

Fig. 7. Spoofing: Dynamic accuracy/loss curves with UAVAttack dataset using 12 clients.

TABLE II
PERFORMANCE METRICS WITH 10 CLIENTS USING CICIoTDIAD2024 DATASET

Traffic	Precision	Recall	F1-score	Accuracy	MAML
Benign	0.96	0.99	0.97	0.9559	0.1599
Attack	0.91	0.77	0.83		

TABLE III
PERFORMANCE METRICS WITH 5 CLIENTS USING CICIDS2017 DATASET

Traffic	Precision	Recall	F1-score	Accuracy	MAML
DDoS	1.00	1.00	1.00	0.9952	0.0533
PortScan	1.00	0.99	1.00		
Infiltration	1.00	0.50	0.67		
WebAttacks	0.97	1.00	0.99		

TABLE IV
PERFORMANCE METRICS WITH 10 CLIENTS USING CICIDS2017 DATASET

Class	Precision	Recall	F1-score	Accuracy	MAML
DDoS	1.00	1.00	1.00	0.9976	0.0420
PortScan	1.00	1.00	1.00		
Infiltration	1.00	0.33	0.50		
WebAttacks	0.98	1.00	0.99		

and attack traffic. The MAML value is 0.1599, indicating the model's capacity to swiftly adapt to new tasks. Table III displays data on the intrusion detection system's performance with 5 clients from the CICIDS2017 dataset. The model scored 99.52% accuracy for most attack types, including DDoS, PortScan, and WebAttacks. Infiltration attacks performed poorly, with a recall of 0.50. This MAML score of 0.0533 shows excellent adaptability to novel attack patterns.

Table IV provides a summary of the model's performance across 10 clients using the CICIDS2017 dataset. The overall accuracy was 99.76%, and the MAML score of 0.0420 indicates great adaptation to changing data on clients. Figure V shows the CICIoV2024 results for three clients, achieving perfect spoofing detection and high scores for DoS and benign traffic. A MAML score of 0.0635 indicates strong adaptability with minimal training, and the overall accuracy is 98.92%.

The performance results of the CICIoV2024 dataset with seven clients are presented in Table VI. The model scored

TABLE V
PERFORMANCE METRICS WITH 3 CLIENTS USING CICIoV2024 DATASET

Class	Precision	Recall	F1-score	Accuracy	MAML
Benign	0.99	0.96	0.97	0.9892	0.0635
DoS	0.97	1.00	0.98		
Spoofing	1.00	1.00	1.00		

TABLE VI
PERFORMANCE METRICS WITH 7 CLIENTS USING CICIoV2024 DATASET

Class	Precision	Recall	F1-score	Accuracy	MAML
Benign	1.00	0.95	0.98	0.9909	0.0849
DoS	0.97	1.00	0.98		
Spoofing	1.00	1.00	1.00		

TABLE VII
PERFORMANCE METRICS WITH 15 CLIENTS USING CICIoV2024 DATASET

Class	Precision	Recall	F1-score	Accuracy	MAML
DoS	0.99	0.99	0.99	0.9960	0.0591
GAS	1.00	0.98	0.99		
RPM	0.98	1.00	0.99		
Speed	0.99	0.99	0.99		
Str: Wheel	0.98	0.99	0.99		

TABLE VIII
COMPARISON OF PERFORMANCE METRICS USING UAVAttack DATASET WITH VARYING NUMBERS OF CLIENTS

Class	Precision	Recall	F1-score	Accuracy	MAML
10 clients					
Normal	1.00	1.00	1.00	0.9965	0.0832
Anomaly	0.99	0.98	0.98		
12 clients					
Normal	1.00	1.00	1.00	0.9986	0.0213
Anomaly	0.99	0.99	0.99		
22 clients					
Normal	1.00	1.00	1.00	0.9959	0.0238
Anomaly	1.00	0.97	0.98		

TABLE IX
PERFORMANCE COMPARISON WITH RELATED PUBLISHED STUDIES

Work	Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Hu et al. [13]	Federated Meta-learning	86.07	–	–	84.00
Yan et al. [19]	Meta-learning IDS	93.10	–	93.5	92.80
Ullah et al. [31]	Federated Edge Intelligence	97.45	98.10	97.75	97.94
Kumar et al. [32]	Federated Active Meta-learning	96.70	96.50	97.20	95.8
He et al. [33]	Federated IDS	84.00	89.03	78.96	83.69
Proposed Approach	**ZTID-IoV**	**99.86**	**100**	**99.00**	**100**

TABLE X
PERFORMANCE VERSUS CONVERGENCE THRESHOLDS

ϵ	Rounds	Accuracy (%)	F1-score (Anomaly) (%)
0.1	10	0.9850	0.96
0.01	35	0.9986	0.99
0.001	60+	0.9988	0.99

TABLE XI
HYPERPARAMETER ABLATION STUDY

Parameter	Values	F1-Score (%)	Rounds
d_token	16	97.2	48
	32	**99.0**	**37**
	64	99.2	42
h	2	98.3	45
	4	**99.0**	**40**
	8	99.1	43
L	1	96.9	50
	2	**99.0**	**42**
	4	99.0	44

99.09% accuracy, with excellent precision for benign and spoofing classes and high overall scores. Table VII shows the CICIoV2024 results for 15 clients, with precision, recall, and F1-scores near 0.99 for DoS, GAS, RPM, Speed, and Steering Wheel attacks. The system achieved 99.60% accuracy and a MAML score of 0.0591, indicating strong adaptation and generalization across diverse client data. Table VIII compares the model's performance on the UAVAttack dataset for 10, 12, and 22 clients. The model achieved high precision, recall, and F1-scores in both normal and anomalous classes. The accuracy ranged from 99.59% to 99.86%. The MAML scores exhibit a slight decrease (by 0.0213 and 0.0238) as the number of clients increases, suggesting that the generalization is improved. This means that the model adapts better to heterogeneous and dispersed data settings, handling variances over a wider number of clients more efficiently. Table IX compares the proposed ZTID-IoV technique with recent federated and meta-learning-based intrusion detection approaches. The comparison includes several performance metrics, such as accuracy, precision, recall, and F1-score. Some prior studies only reported partial metrics, represented by "–" in the table. The suggested method outperforms the baselines in all reported measures, with an accuracy of 99.86%, precision and F1-score of 100%, and recall of 99.00%. This demonstrates its exceptional performance in detecting attacks within the IoVs framework.

The convergence threshold $\epsilon = 0.01$ was empirically optimized using the UAVAttack dataset. Table X illustrates that this value maintains training efficiency while achieving optimal accuracy and F1-score (0.9986 and 0.99), respectively. Using more stringent thresholds ($\epsilon = 0.001$) raises the computing cost but produces minimal improvements. The results indicate that $\epsilon = 0.01$ achieves an optimal balance between model performance and training efficiency. The ablation study is shown in Table XI. The lightweight configuration (`d_token`=32, `h`=4, `L`=2) optimizes performance and efficiency, attaining 99.0% F1-score with the fastest convergence (37–42 rounds). Larger values result in marginal benefits ($\leq$0.2% F1) but slower convergence, whereas smaller values reduce accuracy (0.7–2.1%) despite requiring more rounds. This validates our design selections for the proposed approach.

V. CONCLUSION

This study introduced the neurosymbolic AI framework ZTID-IoV to address growing cybersecurity vulnerabilities in smart consumer environments and IoVs. The application of SOM facilitates interpretable client clustering and enhances model aggregation by grouping behaviorally similar devices. Experimental evaluations are conducted using a diverse array of real-world intrusion datasets, including CICIoTDIAD2024, CICIDS2017, CICIoV2024, and UAVAttack. The model typically achieved an accuracy of 95.59% to 99.86% for essential classes such as DDoS, DoS, Spoofing, and WebAttacks, with F1-scores near 1.00. MAML scores demonstrated exceptional adaptability across client distributions, in support of the model's generalization to decentralized environments. ZTID-IoV improves zero-trust security in next-generation IoV systems and offers a scalable and reliable solution for AI-powered cybersecurity centers.

The proposed approach performs well; however, it may have some research limitations. These limitations may include performance degradation due to extreme data heterogeneity or restricted computational resources at edge devices. Real-time deployment, privacy-preserving aggregation, and adversarial attack robustness may limit its applicability. Future research could integrate multimodal data, use blockchain technology to strengthen trust, and validate the framework in large-scale real-world IoV scenarios to improve its applicability.

REFERENCES

[1] P. Rani et al., "Federated learning-based misbehavior detection for the 5G-enabled Internet of Vehicles," *IEEE Trans. Consum. Electron.*, vol. 70, no. 2, pp. 4656–4664, May 2024.

[2] Q. Lai et al., "Improved transformer-based privacy-preserving architecture for intrusion detection in secure V2X communications," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1810–1820, Feb. 2024.

[3] T. Saranya, S. Sridevi, C. Deisy, T. D. Chung, and M. K. A. A. Khan, "Performance analysis of machine learning algorithms in intrusion detection system: A review," *Proc. Comput. Sci.*, vol. 171, pp. 1251–1260, Jun. 2020.

[4] S. K. Datta, J. Haerri, C. Bonnet, and R. Ferreira Da Costa, "Vehicles as connected resources: Opportunities and challenges for the future," *IEEE Veh. Technol. Mag.*, vol. 12, no. 2, pp. 26–35, Jun. 2017.

[5] S. Safavat and D. B. Rawat, "Asynchronous federated learning for intrusion detection in vehicular cyber-physical systems," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPS)*, May 2023, pp. 1–6.

[6] S. A. Abdel Hakeem and H. Kim, "Advancing intrusion detection in V2X networks: A comprehensive survey on machine learning, federated learning, and edge AI for V2X security," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 8, pp. 11137–11205, Aug. 2025.

[7] X. Qu et al., "A survey on the development of self-organizing maps for unsupervised intrusion detection," *Mobile Netw. Appl.*, vol. 26, no. 2, pp. 808–829, Apr. 2021.

[8] C. Lu, X. Wang, A. Yang, Y. Liu, and Z. Dong, "A few-shot-based model-agnostic meta-learning for intrusion detection in security of Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21309–21321, Dec. 2023.

[9] Y. Chen, N. Ashizawa, C. K. Yeo, N. Yanai, and S. Yean, "Multi-scale self-organizing map assisted deep autoencoding Gaussian mixture model for unsupervised intrusion detection," *Knowledge-Based Syst.*, vol. 224, Sep. 2021, Art. no. 107086.

[10] M. Almi'ani, A. A. Ghazleh, A. Al-Rahayfeh, and A. Razaque, "Intelligent intrusion detection system using clustered self organized map," in *Proc. 5th Int. Conf. Softw. Defined Syst. (SDS)*, Apr. 2018, pp. 138–144.

[11] Z. Wang, Y. Lu, W. Wu, Y. Lu, and H. Wang, "A few-shot and anti-forgetting network intrusion detection system based on online meta learning," in *Proc. IEEE Global Commun. Conf.*, Dec. 2024, pp. 3877–3882.

[12] X. Gong, X. Chen, Z. Zhong, and W. Chen, "Enhanced few-shot learning for intrusion detection in railway video surveillance," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 11301–11313, Aug. 2022.

[13] Y. Hu, J. Wu, G. Li, J. Li, and J. Cheng, "Privacy-preserving few-shot traffic detection against advanced persistent threats via federated meta learning," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 3, pp. 2549–2560, May 2024.

[14] V. Mothukuri, P. Khare, R. M. Parizi, S. Pouriyeh, A. Dehghantanha, and G. Srivastava, "Federated-learning-based anomaly detection for IoT security attacks," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 2545–2554, Feb. 2022.

[15] K. Huang, R. Xian, M. Xian, H. Wang, and L. Ni, "A comprehensive intrusion detection method for the Internet of Vehicles based on federated learning architecture," *Comput. Secur.*, vol. 147, Dec. 2024, Art. no. 104067.

[16] J. Chen, Y. Zhao, Q. Li, X. Feng, and K. Xu, "FedDef: Defense against gradient leakage in federated learning-based network intrusion detection systems," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 4561–4576, 2023.

[17] W. Lu et al., "Stones from other hills: Intrusion detection in statistical heterogeneous IoT by self-labeled personalized federated learning," *IEEE Internet Things J.*, vol. 12, no. 10, pp. 14348–14361, May 2025.

[18] M. P. Uddin, Y. Xiang, M. Hasan, J. Bai, Y. Zhao, and L. Gao, "A systematic literature review of robust federated learning: Issues, solutions, and future research directions," *ACM Comput. Surveys*, vol. 57, no. 10, pp. 1–62, Oct. 2025.

[19] H. Yan, X. Lin, S. Li, H. Peng, and B. Zhang, "Global or local adaptation? Client-sampled federated meta-learning for personalized IoT intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 279–293, 2024.

[20] M. Rabbani et al., "Device identification and anomaly detection in IoT environments," *IEEE Internet Things J.*, vol. 12, no. 10, pp. 13625–13643, May 2025.

[21] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," *ICISSp*, pp. 108–116, 2018.

[22] E. C. P. Neto et al., "CICIoV2024: Advancing realistic IDS approaches against DoS and spoofing attack in IoV can bus," *Internet Things*, vol. 26, Jul. 2024, Art. no. 101209.

[23] J. Whelan, T. Sangarapillai, O. Minawi, A. Almehmadi, and K. El-Khatib, "Novelty-based intrusion detection of sensor attacks on unmanned aerial vehicles," in *Proc. 16th ACM Symp. QoS Secur. Wireless Mobile Netw.*, Nov. 2020, pp. 23–28.

[24] J. J. Davis and A. J. Clark, "Data preprocessing for anomaly based network intrusion detection: A review," *Comput. Secur.*, vol. 30, nos. 6–7, pp. 353–375, Sep. 2011.

[25] S. Ullah et al., "TNN-IDS: Transformer neural network-based intrusion detection system for MQTT-enabled IoT networks," *Comput. Netw.*, vol. 237, Dec. 2023, Art. no. 110072.

[26] Y. Lu, X. Huang, Y. Dai, S. Maharjan, and Y. Zhang, "Federated learning for data privacy preservation in vehicular cyber-physical systems," *IEEE Netw.*, vol. 34, no. 3, pp. 50–56, May 2020.

[27] Z. Liu, J. Guo, W. Yang, J. Fan, K.-Y. Lam, and J. Zhao, "Privacy-preserving aggregation in federated learning: A survey," *IEEE Trans. Big Data*, early access, Jul. 15, 2022, doi: 10.1109/TBDATA.2022.3190835.

[28] A. Yang et al., "Application of meta-learning in cyberspace security: A survey," *Digit. Commun. Netw.*, vol. 9, no. 1, pp. 67–78, Feb. 2023.

[29] R. Wehrens and J. Kruisselbrink, "Flexible self-organizing maps in Kohonen 3.0," *J. Stat. Softw.*, vol. 87, no. 7, pp. 1–18, 2018.

[30] S. Dey, Q. Ye, and S. Sampalli, "A machine learning based intrusion detection scheme for data fusion in mobile clouds involving heterogeneous client networks," *Inf. Fusion*, vol. 49, pp. 205–215, Sep. 2019.

[31] F. Ullah, L. Mostarda, D. Cacciagrano, M. J. F. Alenazi, C.-M. Chen, and S. Kumari, "Federated edge intelligence for enhanced security in consumer intermittent healthcare devices using adversarial examples," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 4574–4585, May 2025.

[32] K. Kumar and M. Khari, "Federated active meta-learning with blockchain for zero-day attack detection in industrial IoT," *Peer-Peer Netw. Appl.*, vol. 18, no. 4, pp. 1–47, Jul. 2025.

[33] S. He, C. Du, and M. S. Hossain, "6G-enabled consumer electronics device intrusion detection with federated meta-learning and digital twins in a meta-verse environment," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3111–3119, Feb. 2024.